RO-Crate: Capturing FAIR research outputs throughout the ELIXIR landscape

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RO-Crate

Overview

RO-Crate is an **open, community-driven, lightweight** approach to packaging research objects with their metadata.

Using RO-Crate makes research outputs more **attributable**, **interconnected**, and **FAIR**.

Benefits of RO-Crate

RO-Crate uses **linked data**, which means RO-Crate metadata is naturally interconnected, interoperable, and queryable.

The **RO-Crate specification** is general enough to be usable in many different infrastructures and domains, and to describe all kinds of research objects.

Custom **RO-Crate profiles** extend the specification to meet the needs of a particular domain or community (examples below).

RO-Crate enables **credit and attribution** for all contributors, including those in currently under-recognised roles (e.g. sample gathering, processing, sample distribution).

Workflows

WorkflowHub provides an RO-Crate download for every registered workflow, using the *Workflow RO-Crate* profile.

Several workflow engines and orchestrators, including Galaxy, WfExS, CWL, and Nextflow, export RO-Crates following the *Workflow Run Crate* profile, which captures information about a specific execution of a workflow.

Interactive Example

RO-Crates can be rendered into human-friendly web pages! Browse an example:

https://bit.ly/ro-crate-example

Bioinformatics Projects

Publishing simulations of wild relatives of crops as *Workflow Run Crates*

Capturing complex plant science experiments using the *Annotated Research Context (ARC)* profile

Defining an *RO-Crate profile for biodiversity genomics*

COPO platform can export genomic data as RO-Crates

Other ELIXIR Projects

Data Archiving

RO-Crate is suitable for long-term data archival due to its clearly-defined specification and use of well-supported Web technologies. It can be used with any type of data.

ROHub is a data repository that uses RO-Crate natively. Other repositories can also implement support for importing and exporting RO-Crates - as in Dataverse and FAIRDOM-SEEK.

TREs

Five Safes RO-Crate captures provenance of workflow executions within Trusted Research Environments (TREs).

Includes processes like sign-off, workflow execution, and disclosure control, without requiring sensitive data to be published.

RO-Crate Community

RO-Crate is developed in the open on GitHub[1]. Anyone can join our global community to discuss applications, profiles, and tools.

Start with attending a beginner-friendly **drop-in session**[2]!

- **First Thursday of the month, 9am UK time**
- **Third Wednesday of the month, 3pm UK time**

To learn more about RO-Crate, visit https://w3id.org/ro/crate/

References:
1. https://github.com/ResearchObject/ro-crate
2. https://s.apache.org/ro-crate-regional

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